Frank Santaguida

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Skills

AI Engineering: Python | OpenAI | RAG | MCP | LangChain | Pinecone | FastAPI
Full-Stack & Infra: JavaScript, TypeScript | Next.js | React | AWS | SQL, GraphQL, NoSQL

Experience

AI Software Engineer | Penn State University | PA | Feb 2023– May 2026

- Built **MaterialsTube**, an **NSF**-funded video platform for the national materials science community
- Delivered a full **RAG** pipeline for handling video data and facilitating semantic searches using **AWS**, **Pinecone** vector store, **OpenAI embeddings**, **LangChain** and **Completions API**
- Engineered the LiST agentic chat interface using a ReAct-style agent loop and **Responses API**, deriving dynamic tool schemas directly from a production **OpenAPI** spec to enforce strict safety constraints
- Implemented a **FastMCP** server, providing access to **LiST**'s API via **MCP** protocol
- Rewrote legacy **Knockout.js** application into a modern **Next.js** / **React** / **TypeScript** SPA
- Deployed and maintained serverless AWS infrastructure — including **Lambda, DynamoDB, S3, CloudFormation, Cognito** and **GraphQL** — to orchestrate AI video transcriptions, vector processing, and complex data pipelines.
- Built LLM-as-judge evaluators, benchmarking agent and model responses for correctness, speed, cost
- Developed a **Next.js** / **FastAPI** hybrid service, handling scientific data processing, such as AFM image normalization, XRD peak detection, and **D3** visualizations for multi-modal materials characterization data
- Ensured all deliverables met **ADA** and **WCAG** accessibility standards across platforms
- Led internal AI education initiatives, defining and translating technical concepts into presentations
- Served on the Materials Research Institute Staff Advisory Committee

Web Developer | 1SEO Digital Agency | PA | 2022 – 2023

- Developed and maintained client-facing websites using **TypeScript**, **React**, and **React Router**
- Built and optimized high-volume websites for local service businesses using **Google Analytics** and CRM integrations, improving page load performance and SEO rankings
- Acted as the technical lead for key accounts, translating complex client business requirements into actionable engineering scopes and delivering custom front-end solutions

Engineer / Consultant | Built by Frank | Philadelphia, PA | 2012 – 2022

- Led an international development team through agile workflows, managing full project lifecycles spanning database design, security implementation, and end-to-end full-stack delivery.
- Consulted on product roadmaps and delivered technical presentations to enterprise stakeholders and investors to align engineering solutions with business objectives.
- Developed **Progressive Web App (PWA)** to manage complex real estate continuing education credentials
- Architected custom **Zoho CRM** schemas and developed robust data pipelines to map, validate, and consume web-scraped data and **RESTful APIs** across web applications.
- Deployed and managed scalable cloud infrastructure utilizing **Heroku** and **Netlify**, building automated **CI/CD** pipelines to ensure high availability and streamlined deployment cycles
- Architected complex third-party enterprise integrations — including **Cloudflare**, **Google Maps API**, and **WooCommerce** — managing webhooks, authentication, and secure data synchronization
- Managed client enablement through hands-on technical training, system handover sessions, and proactive reporting to ensure on-time project delivery.

Market Operations Consultant | Bravosolution, now Jaggaer | Malvern, PA | 2011

- Optimized sourcing and spend practices, capturing millions of dollars and increasing profit through visibility
- Managed **RFQs, B2B** auctions, and supplier-client relations, selling millions in construction equipment
- Led implementation of help-desk process changes, technical documentation, and reporting

ERP & Accounting Specialist | Deacom | Wayne, PA | 2010

- Supported DEACOM **ERP** software and resolved accounting discrepancies involving general ledgers
- Resolved critical "production down" emergencies within strict SLA constraints
- Administered **Microsoft SQL Server** configurations, backups, and restorations

Associate ERP Support Analyst | Infor | Malvern, PA | 2008 – 2010

- Supported Adage, MyDay ERP software, as direct point of contact for major clients
- Engineered custom **SQL** scripts to resolve emergency client database issues
- Provisioned isolated staging environments with virtual machines, **Apache Tomcat**, and **SQL**
- Led the "Green Beans" graduate organization, authored technical documentation

Researcher & Developer | Pennsylvania State University | State College, PA | 2008

The Network-Centric Cognition and Information Fusion Center

- Designed prototype UI for Penn State's security command center using MVC
- Developed data-driven security solutions and proposals for the university campus
- Built and populated a technical lexicon for the NC2IF center using Wikidot

Education

BS in Information Sciences & Technology | Pennsylvania State University | 2008